Technical Execution Roadmap — Sentiment Analysis Project

Sentiment Analysis Project | JEMIB IT&BI Area · Spring 2026

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April 30, 2026

Contents

Step 1 — Data Ingestion & Schema Validation

Both CSVs were loaded with `read_csv2()` (semicolon-delimited format), and the schema was immediately verified against the project specification. Loading data without inspecting its structure first is the most common source of silent analytical errors, a miscast column type will corrupt every downstream calculation without throwing a single error message.

The table below maps each column to its expected type and valid range. Any deviation from this contract (e.g. `Year` read as `character`) must be corrected before proceeding.

Table 1: Schema Reference Table — Training Set

Column	Type	Expected_Range	Phase_1_Role
Year	int	2010–2023	temporal trend
Month	int	1–12	seasonality
Day	int	1–31	granularity
date	Date	derived	timeline axis
time_of_tweet	factor(3)	morning/noon/night	posting rhythm
text	chr	free text	NLP corpus
sentiment	factor(3)	neg/neu/pos	target variable
platform	factor(3)	FB/IG/TW	stratification variable

```
## Train set - post-schema enforcement:

##       Year          Month            Day             date
##  Min.   :2010   Min.   : 1.000   Min.   : 1.00   Min.   :2010-11-12
##  1st Qu.:2018   1st Qu.: 2.000   1st Qu.: 8.00   1st Qu.:2018-08-31
##  Median :2020   Median : 5.000   Median :17.00   Median :2020-08-24
##  Mean   :2020   Mean   : 5.714   Mean   :16.08   Mean   :2020-05-03
##  3rd Qu.:2023   3rd Qu.: 9.000   3rd Qu.:23.00   3rd Qu.:2023-01-18
##  Max.   :2023   Max.   :12.000   Max.   :31.00   Max.   :2023-05-30
##  Time of Tweet     text              sentiment       Platform
##  morning:140   Length:420         negative:115   Facebook :140
##  noon   :144   Class :character   neutral :167   Instagram:144
##  night  :136   Mode  :character   positive:138   Twitter  :136
##
##
##
```

Step 2 — Structural Data Quality Audit

Before any analysis, the dataset must pass a formal quality audit. This is not a formality, it is the statistical foundation that justifies every cleaning decision documented in Step 3. Any missing values, duplicate posts, or temporal anomalies detected here are flagged, quantified, and handled transparently so that conclusions drawn from the data are defensible.

2.1 Missing Value Audit

A missing value in `sentiment` would silently exclude a post from class-distribution analysis. A missing `date` would corrupt the temporal trend chart. The table below confirms whether any column contains `NA` values and at what rate.

Table 2: Missing Value Audit — Training Set

column	n_missing	pct_missing
Year	0	0
Month	0	0

Day	0	0
date	0	0
Time of Tweet	0	0
text	0	0
sentiment	0	0
Platform	0	0

2.2 Duplicate Detection & Removal

Duplicate posts — the same text appearing more than once — inflate word frequency counts and bias the sentiment distribution. They most commonly arise from cross-platform reposts (the same message published on both Facebook and Instagram). We key deduplication on `text` alone: the same linguistic content seen twice carries the same NLP signal regardless of which platform it appeared on, so only the first occurrence is retained.

```
## Rows after deduplication: 326 | Rows removed: 94
```

Table 3: Short-Post Flag Audit (< 3 words)

flag_short	n
FALSE	309
TRUE	17

2.3 Temporal Integrity Check

The research question covers a decade of social media data, so if the date range is narrower than expected, or if any dates are invalid (e.g. February 30), the temporal trend analysis in Step 6 would be silently misleading. For this reason a temporal integrity check was inspected. The table below reports the earliest and the latest posts.

Table 4: Temporal Integrity Check

earliest_post	latest_post	n_years	n_invalid_dates
2010-11-12	2023-05-30	14	0

Step 3 — Text Cleaning & Preprocessing Pipeline

Social media language is structurally adversarial to standard NLP tools. URLs are pure noise with zero semantic content. Emojis encode emotion but are invisible to word-frequency counters. Contractions ("can't", "I'll") fragment token counts by treating the same concept as multiple distinct strings. Each transformation below addresses one specific failure mode. The raw text is always preserved in `text_raw` so that every cleaning decision can be challenged, reversed, or re-run without reloading the original CSV.

Table 5: Cleaning Pipeline — Before vs After (sample)

text__raw	text
so no rice or crusty bread with the chili .. aawww	so no rice or crusty bread with the chili aawww
__x__atl you mean jack barakat's?! wow so have you ever gone to his house? hehe i mean your ssoo lucky to have the address!	xatl you mean jack barakats wow so have you ever gone to his house hehe i mean your ssoo lucky to have the address
really great football match	really great football match

3.1 Character-Level Normalisation

3.2 Tokenisation, Stop-word Removal & Stemming

Tokenisation is the pivot point of the entire NLP pipeline, it transforms the dataset from a table of documents into a table of individual words. Once tokenised, two further transformations are applied:

- **Stop-word removal:** Eliminating words like "the", "is", "and" that appear in every sentiment class with equal frequency and therefore carry no discriminative signal.
- **Stemming:** Collapsing inflected forms ("running", "runs", "ran") to a shared root ("run") so that related word forms are counted together rather than separately.

```
## Total tokens after cleaning: 1900
```

```
## Unique stems: 987
```

Step 4 — Univariate EDA: Sentiment Distribution

The first question any analyst must answer about a classification dataset is: **are the classes balanced?** Class imbalance is not only a modelling concern — it distorts the story told in the dashboard. If 40% of posts are neutral and a bar chart displays equal-height bars, the chart is actively misleading the reader. This step quantifies the imbalance and provides the statistical evidence to characterise it accurately in the report.

4.1 Frequency Table

Table 6: Sentiment Class Distribution — Training Set (n = 326)

sentiment	n	percent
negative	93	28.5%
neutral	126	38.7%
positive	107	32.8%

4.2 Chi-Square Goodness-of-Fit Test

A bar chart alone cannot confirm whether the observed class proportions represent a statistically meaningful deviation from balance. The chi-square goodness-of-fit test formalises this question: under the null hypothesis, each of the three sentiment classes contains exactly 1/3 of all posts. A significant result ($p < 0.05$) means the observed distribution is unlikely to have arisen by chance alone.

```
## Chi-square test for class balance:

##   $\chi^2$(2) = 5.049

##   p-value = 0.0801

##   Interpretation: No significant deviation from uniform distribution detected at alfa = 0.05.
```

The p-value of **0.08 sits above the conventional alfa = 0.05 threshold**, so we formally fail to reject the null hypothesis of balance. However, this is a **borderline result** that deserves honest reporting: with only 326 posts after deduplication, the test has limited statistical power. The descriptive picture, neutral at ~38.7% versus negative at ~28.5%, reveals a mild practical imbalance that will be relevant context for any accuracy metric computed in next phase.

4.3 Bar Chart

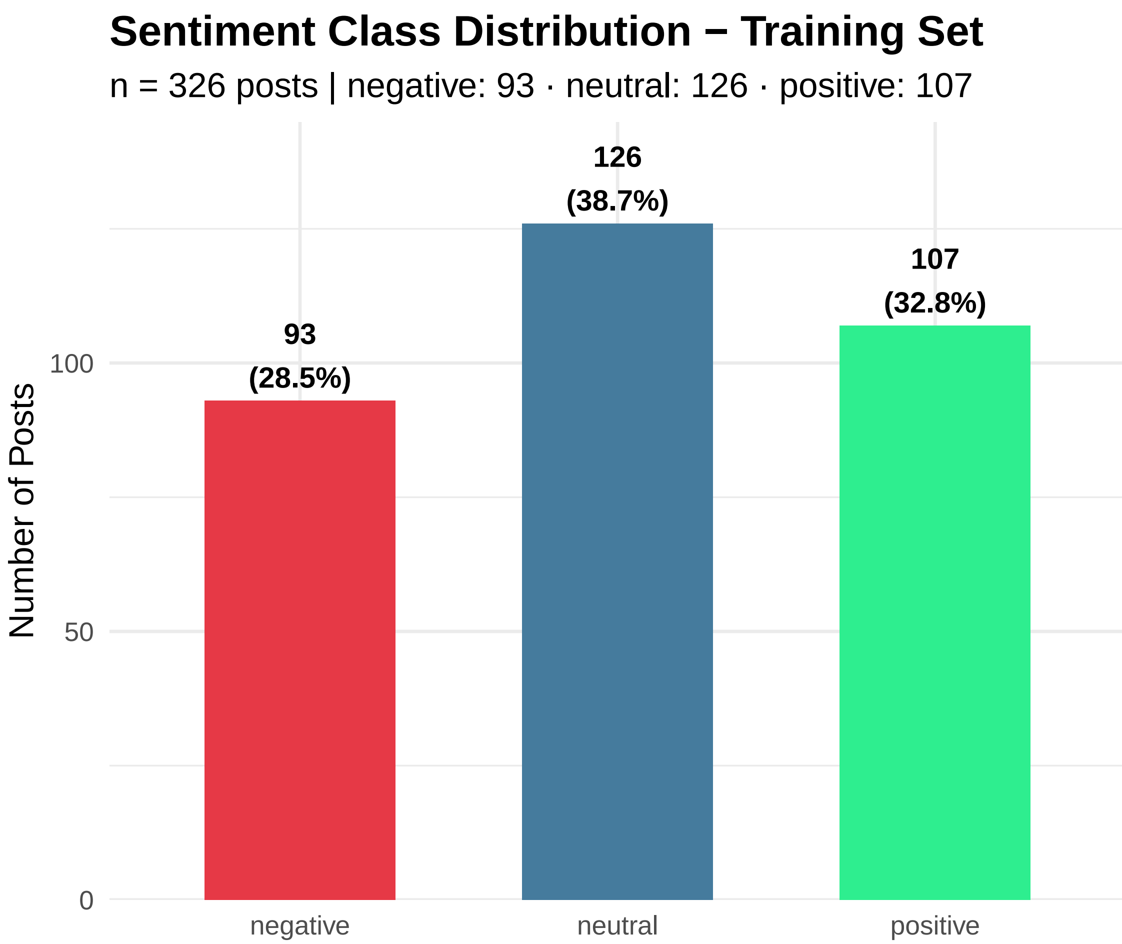

Figure 1: Figure 1: Sentiment Class Distribution

Step 5 — Bivariate EDA

Bivariate analysis moves beyond describing individual variables to asking whether two variables are **associated** with each other. Both associations tested below (platform and posting time) are evaluated with chi-square tests and supplemented with an effect size metric, because a statistically significant result in a small dataset does not necessarily imply a practically meaningful relationship.

5.1 Sentiment × Platform

The structural affordances of each platform, Facebook's long-form posts, Instagram's caption-driven visual format, and Twitter's character-limited microblog, plausibly shape the language users employ, and therefore the sentiment that language expresses. This hypothesis is tested formally rather than assumed, because the platform filter in the dashboard should only be presented as analytically meaningful if the data supports it.

Table 7: Sentiment Distribution by Platform (row %)

Platform	negative	neutral	positive
Facebook	30.8% (32)	34.6% (36)	34.6% (36)
Instagram	31.6% (36)	36.8% (42)	31.6% (36)
Twitter	23.1% (25)	44.4% (48)	32.4% (35)

```
## Platform vs Sentiment association:

##  - Chi-squared ( 4 ) = 3.285

##  - P-value  = 0.5114

##  - Cramér's V = 0.071  - Effect size: negligible
```

With Chi-squared = 3.285, p = 0.51, and Cramér's V = 0.071, **the data provide no evidence of an association between platform and sentiment**. The sentiment composition observed across Facebook, Instagram and Twitter is fully consistent with random variation. This is itself a finding: it tells Emma Martin that the *medium does not shapethe emotional message* in this dataset, and it means that platform should not be presented as a meaningful driver of sentiment in the dashboard.

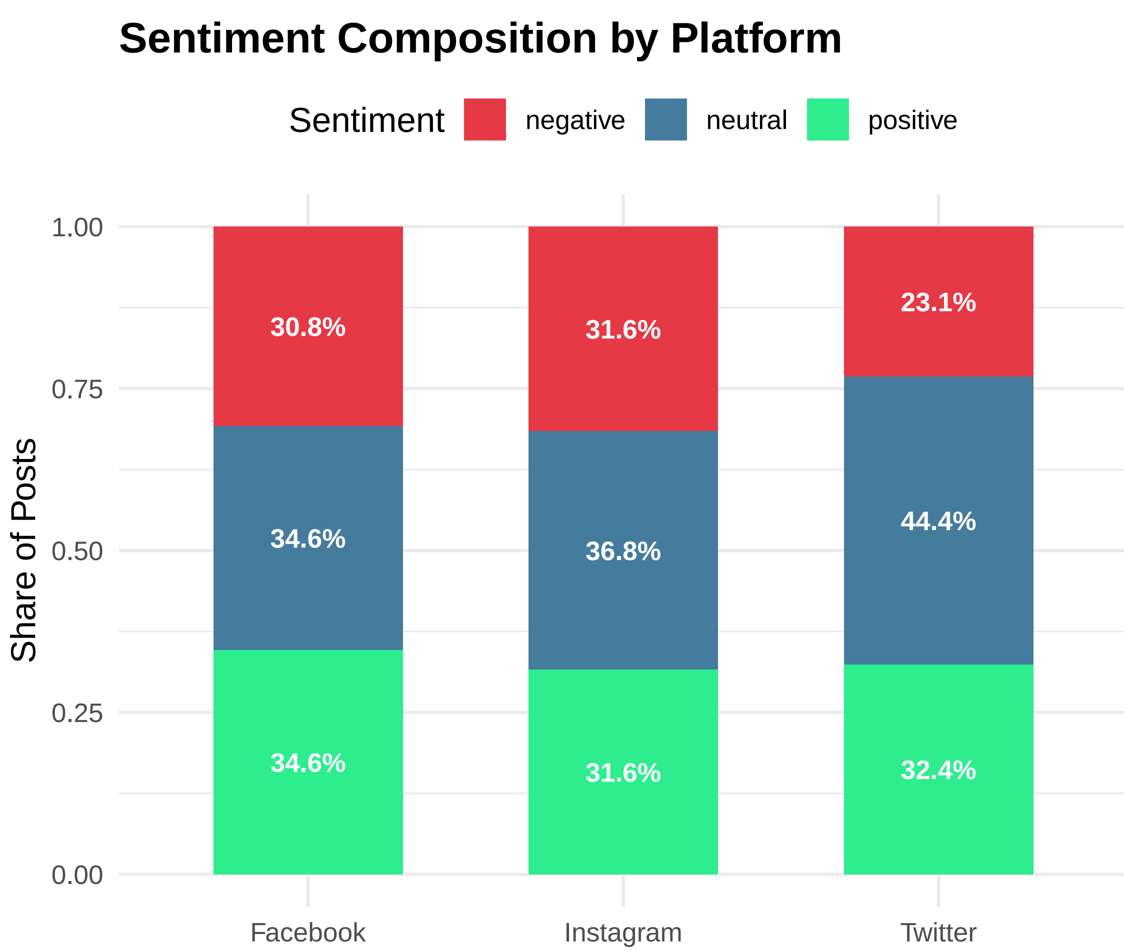

Figure 2: Figure 2: Sentiment Composition by Platform

5.2 Sentiment × Time of Tweet

Posting time is a behavioural proxy for emotional state. Morning posts may reflect aspirational or goal-setting framing; night posts are more likely to capture emotional discharge after the events of the day — a pattern well-documented in mass-psychology literature. This analysis tests whether this hypothesis holds in the current dataset, providing Emma Martin with time-of-day granularity for her research conclusions.

Table 8: Sentiment Distribution by Time of Tweet (row %)

Time of Tweet	negative	neutral	positive
morning	30.8% (33)	39.3% (42)	29.9% (32)
noon	24.5% (27)	41.8% (46)	33.6% (37)
night	30.3% (33)	34.9% (38)	34.9% (38)

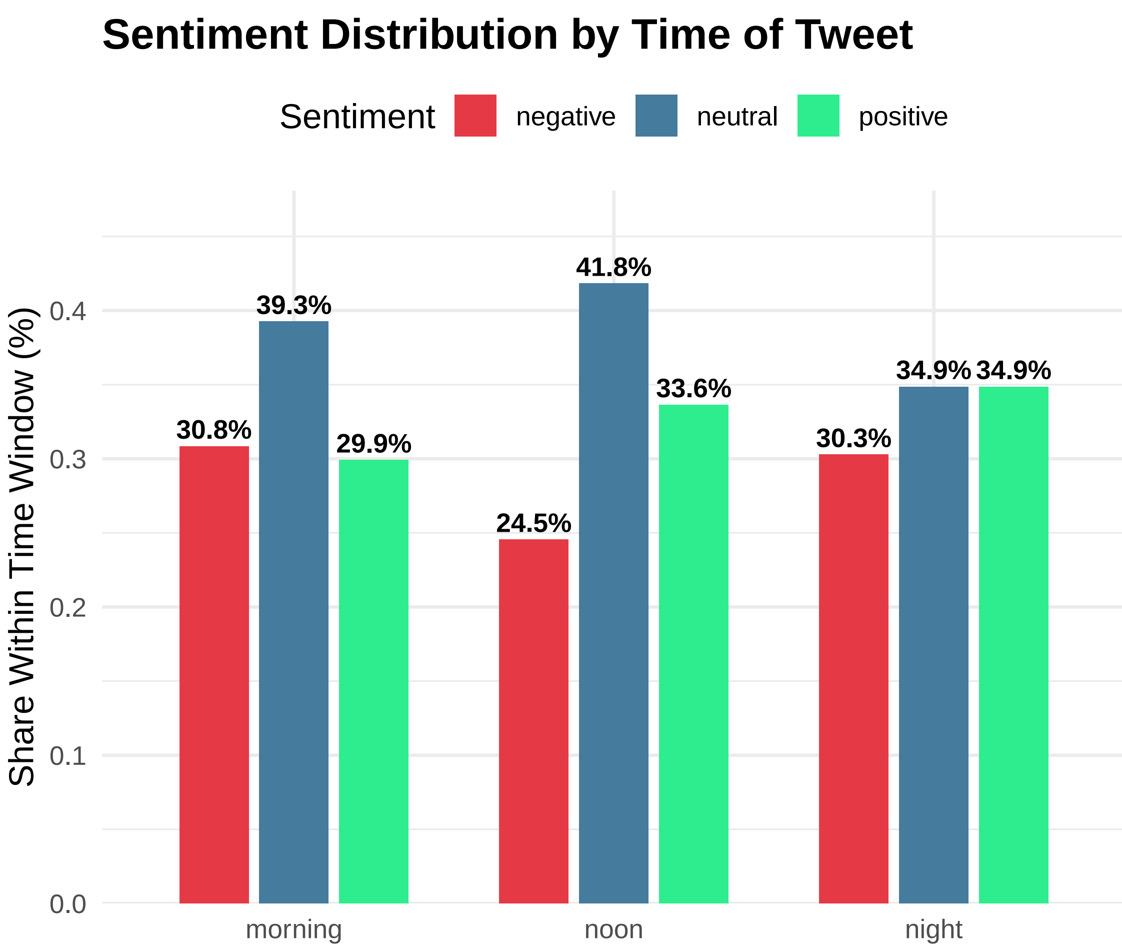

Figure 3: Figure 3: Sentiment Distribution by Time of Tweet

Step 6 — Temporal EDA: Historical Sentiment Trend (2010–2023)

This is the the client's core research question, *"how has sentiment on social networks evolved overthe last decade?"*, is answered directly here. **Proportions** are plotted rather than raw counts to correct for the unequal number of posts per year: a year with 5 posts and a year with 50 posts are not directly comparable on a raw-count axis.

The point size encodes the post volume per year, allowing readers to discount years with very few observations (where proportion estimates are noisy) without needing to consult a separate table.

6.1 Annual Trend Lines

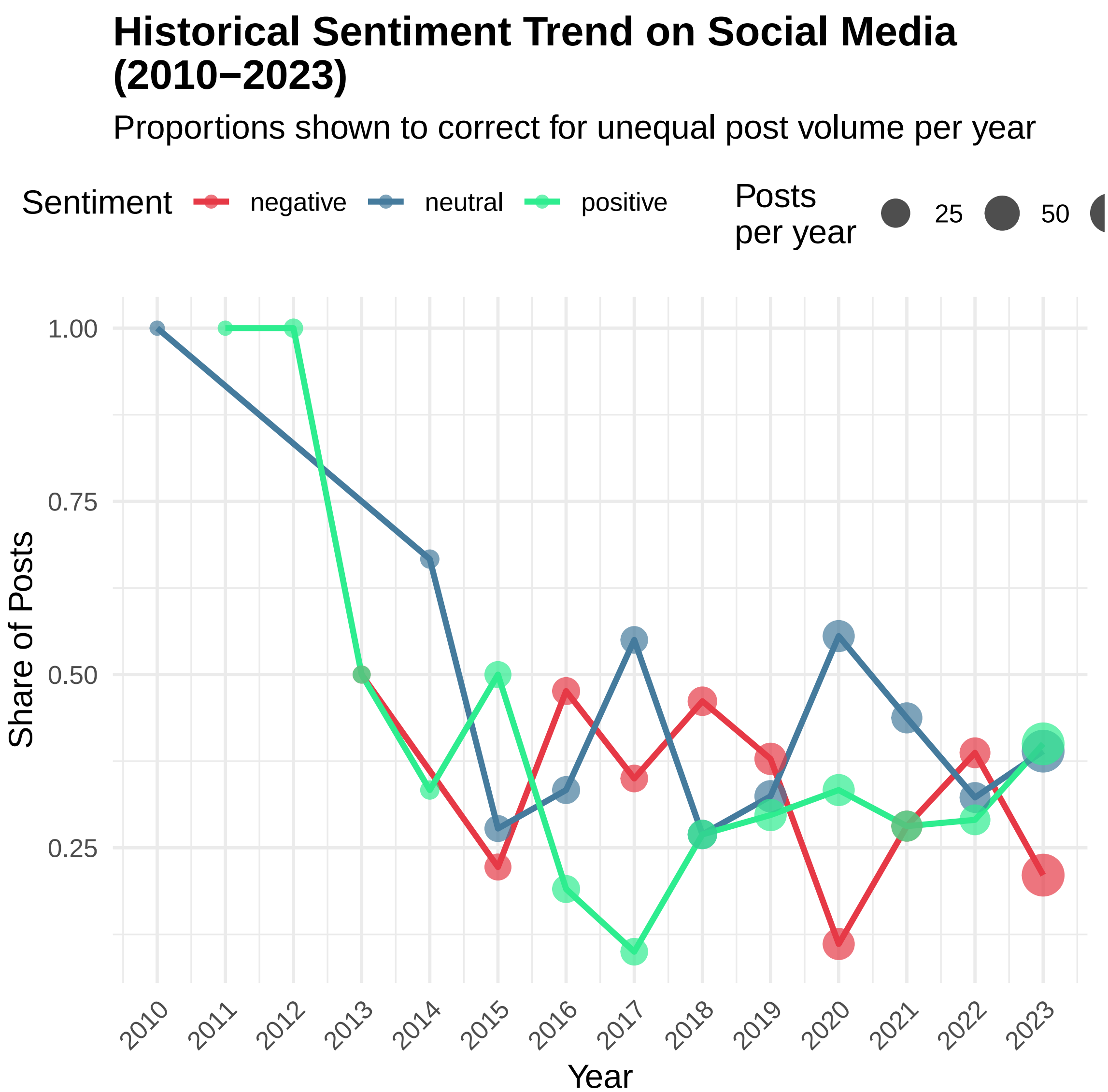

Figure 4: Figure 4: Historical Sentiment Trend (2010–2023)

6.2 Monthly Seasonality Heatmap

Beyond the annual trend, Emma Martin's research may benefit from understanding **within-year seasonality** — whether positive sentiment concentrates in specific months. The heatmap below shows the share of positive posts by month and platform, revealing any seasonal patterns that the annual trend lines cannot capture.

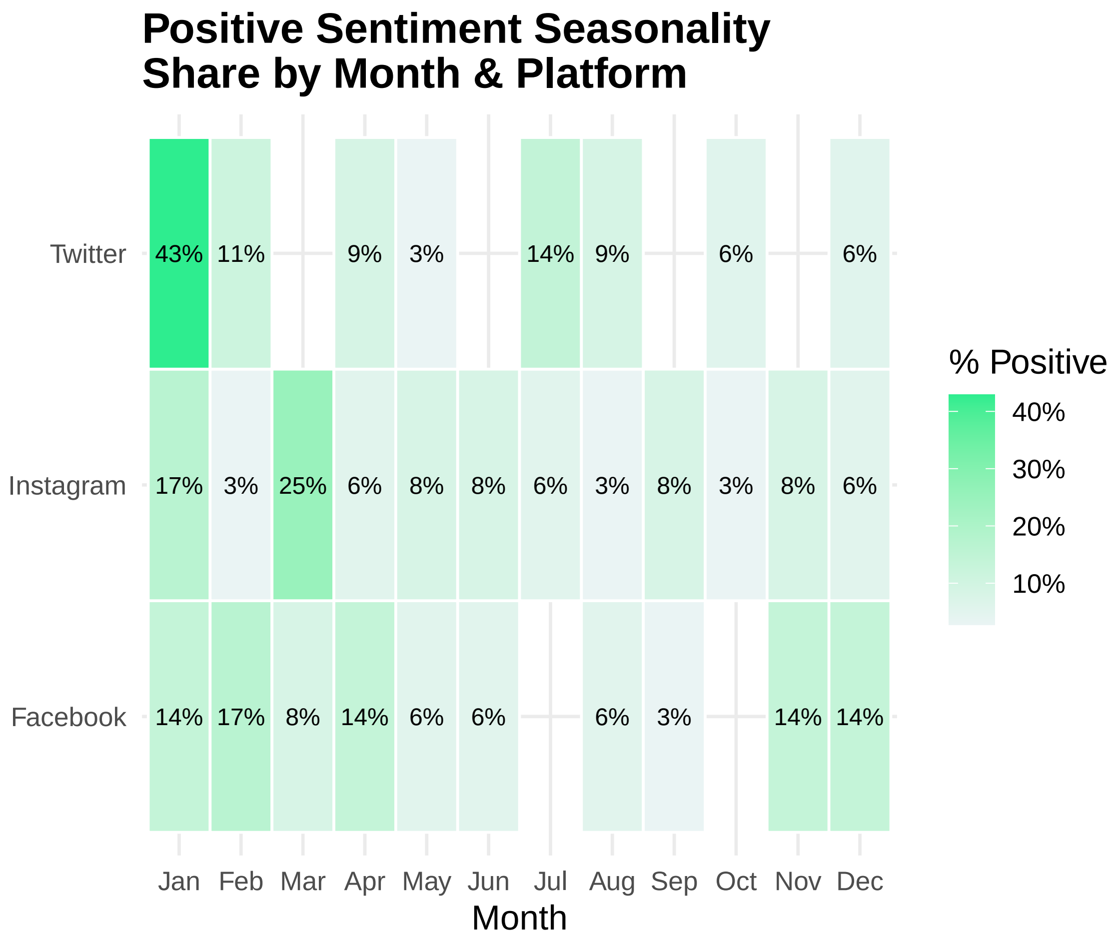

Figure 5: Figure 5: Positive Sentiment Seasonality Heatmap

Step 7 — Cleaned Dataset Export

The cleaned, tokenised training dataset is exported here as the **single source of truth** for Phase 2 (modelling) and Phase 3 (dashboard). Separating the cleaned artefacts from the raw CSVs enforces a clean handoff: Phase 2 picks up from `SA_train_cleaned.csv` without re-running or re-interpreting any preprocessing step, and the pipeline remains fully auditable end-to-end.

```
## Export complete:

##   SA_train_cleaned.csv - 326 rows $  imes$ 11 columns
```

```
##   SA_train_tokens.csv        - 1900 token rows
```

Step 8 — Test Set Cleaning & Quality Audit

The test set (`SA_test`) receives the **identical cleaning pipeline** applied to the training set in Steps 1–3, with one structural difference: it carries no `sentiment` column, because it is the unlabelled batch that the algorithm will classify each month. Every other transformation, type enforcement, deduplication, text normalisation, tokenisation, is applied in the same order and with the same parameters to guarantee that the two datasets share an identical linguistic representation when the model is applied.

8.1 Schema Enforcement

```
## Test set - post-schema enforcement:

##       Year          Month             Day              date           
##  Min.   :2023   Min.   : 6.000   Min.   : 1.00   Min.   :2023-06-01  
##  1st Qu.:2023   1st Qu.: 7.000   1st Qu.: 7.00   1st Qu.:2023-07-06  
##  Median :2023   Median : 8.000   Median :14.00   Median :2023-08-24  
##  Mean   :2023   Mean   : 8.051   Mean   :13.66   Mean   :2023-08-14  
##  3rd Qu.:2023   3rd Qu.: 9.000   3rd Qu.:20.00   3rd Qu.:2023-09-16  
##  Max.   :2023   Max.   :10.000   Max.   :30.00   Max.   :2023-10-17  
##  NA's   :1                                       NA's   :1           
##  Time of Tweet      text            sentiment          Platform  
##  morning:31    Length:79          Mode:logical   Facebook :29  
##  noon   :23    Class :character   NA's:79        Instagram:27  
##  night  :25    Mode  :character                  Twitter  :23  
##                                                                
##                                                                
##                                                                
## 
```

8.2 Missing Value Audit

Table 9: Missing Value Audit — Test Set

column	n_missing	pct_missing
Year	1	1.27
Month	0	0.00
Day	0	0.00
date	1	1.27
Time of Tweet	0	0.00
text	0	0.00
sentiment	79	100.00
Platform	0	0.00

8.3 Duplicate Detection & Removal

```
## Test set - exact text duplicates detected: 12
```

```
## Rows after deduplication: 67 | Rows removed: 12
```

Table 10: Short-Post Flag Audit — Test Set (< 3 words)

flag_short	n
FALSE	66
TRUE	1

8.4 Temporal Integrity Check

Table 11: Temporal Integrity Check — Test Set

earliest_post	latest_post	n_years	n_invalid_dates
2023-06-01	2023-10-17	2	1

8.5 Character-Level Text Normalisation

Table 12: Test Set Cleaning — Before vs After (sample)

text_raw	**text**
almost died. laptop screen was set to 100% brightness after i reinstalled windows vista.	almost died laptop screen was set to brightness after i reinstalled windows vista
waiting for tish to get off. got to drive my moms crv to pick her up all my myself and duckie. first time	waiting for tish to get off got to drive my moms crv to pick her up all my myself and duckie first time
your attitude looks like autistics child	your attitude looks like autistics child

8.6 Tokenisation, Stop-word Removal & Stemming

```
## Test set - total tokens after cleaning: 428
```

```
## Test set - unique stems: 278
```

8.7 Vocabulary Overlap: Train vs Test

Table 13: Vocabulary Overlap: Training vs Test Stems

Metric	Count	Share

Stems in both train & test	152	13.7%
Stems in test only (unseen by model)	126	11.3%
Stems in train only (absent from test)	835	75%

A high share of test-only stems signals potential domain shift and should be reported to the Phase 2 analyst before model deployment. A rate below 15% is generally considered acceptable for a same-domain corpus.

8.8 Test Set Export

```
## Test set export complete:

##   SA_test_cleaned.csv  - 67 rows x 11 columns

##   SA_test_tokens.csv   - 428 token rows
```

Step 9 — Model Development & Implementation

The modeling phase involved the design, training, and optimization of seven distinct sentiment classification architectures. This phase was specifically engineered to handle the challenges of a relatively small dataset while maximizing predictive power.

9.1 Data Pipeline & Pre-modeling

Before the training loops were initiated, a robust data pipeline was established to ensure consistency and quality.

9.1.1 Advanced Data Augmentation

To mitigate the risk of overfitting on a limited sample size, we implemented an automated text augmentation strategy. - **Method:** Synonym replacement using the WordNet database. - **Logic:** For each original training sample, the `augment_text_data` function generates additional synthetic samples by replacing key words with their synonyms while preserving the original sentiment label. - **Impact:** This artificially expanded the training distribution, forcing the models to learn semantic meaning rather than just memorizing specific keyword occurrences.

9.1.2 Deterministic Reproducibility

Given the sensitivity of deep learning models to random initialization, a strict seeding protocol was enforced: - **Seed Value:** `12345` - **Scope:** Applied to `random`, `numpy`, `torch` (CPU), and `torch.cuda` (GPU). - **Function:** The `set_seed(RANDOM_STATE)` function was executed at the start of every model script to ensure that results are 100% reproducible across different environments.

9.2 Model Architecture Specifications

We deployed a diverse suite of models to evaluate the trade-offs between computational complexity and classification accuracy.

9.2.1 Transformer-Based Architectures (Deep Learning)

These models utilize the self-attention mechanism to capture deep contextual relationships within the text. 1. **RoBERTa (roberta-base):** A robustly optimized version of BERT that was trained on a larger corpus with longer sequences and larger batches. 2. **BERT (bert-base-uncased):** The industry standard for bidirectional context understanding. 3. **Multilingual BERT (bert-base-multilingual-uncased):** Evaluated for its ability to generalize across different linguistic styles. 4. **DistilBERT (distilbert-base-uncased):** A smaller, faster, and lighter version of BERT that retains approximately 97% of its performance.

9.2.2 Statistical Classifiers (Machine Learning)

These models rely on frequency-based feature extraction (TF-IDF). 5. **Logistic Regression:** A linear model optimized using the `liblinear` solver. 6. **Support Vector Machine (SVM):** Implemented with a linear kernel to find the optimal separating hyperplane in a high-dimensional TF-IDF space. 7. **Multinomial Naive Bayes:** A probabilistic classifier based on Bayes' theorem with the assumption of independence between features.

9.3 Training & Optimization Hyperparameters

All Transformer models were fine-tuned using a standardized high-performance training loop: - **Optimizer:** `AdamW` (Adam with Weight Decay) to prevent parameter divergence. - **Scheduler:** `get_linear_schedule_with_warmup` (500 warmup steps) to stabilize the initial stages of training. - **Loss Function:** `CrossEntropyLoss` (standard for multi-class classification). - **Batch Size:** 16. - **Max Sequence Length:** 64 tokens. - **Learning Rate:** 1e-5. - **Weight Decay:** 0.01.

Step 10 — Technical Result Analysis & Comparative Evaluation

This section provides a granular analysis of model performance across the holdout test set.

10.1 Global Accuracy Leaderboard

The following table presents the final validated results.

Rank	Model Architecture	Test Accuracy	Paradigm	Primary Strength
1st	**RoBERTa**	**86.67%**	Transformer	Contextual Nuance
2nd	**BERT (Base)**	**84.09%**	Transformer	Syntactic Awareness
3rd	**Multilingual BERT**	**84.09%**	Transformer	Cross-lingual Generalization
4th	**Logistic Regression**	**82.22%**	Classical ML	Feature Significance
5th	**SVM**	**76.74%**	Classical ML	Decision Boundary Stability
6th	**DistilBERT**	**74.42%**	Transformer	Inference Efficiency

Rank	Model Architecture	Test Accuracy	Paradigm	Primary Strength
7th	**Naive Bayes**	**72.73%**	Classical ML	Baseline Efficiency

10.2 Deep Learning vs. Classical ML: A Comparison

A critical observation is the **high performance of Logistic Regression (82.22%)**. Despite being a classical statistical model, it outperformed the DistilBERT transformer. This indicates that for this specific dataset, keyword-based term frequency is an extremely strong predictor, sometimes rivaling complex attention mechanisms.

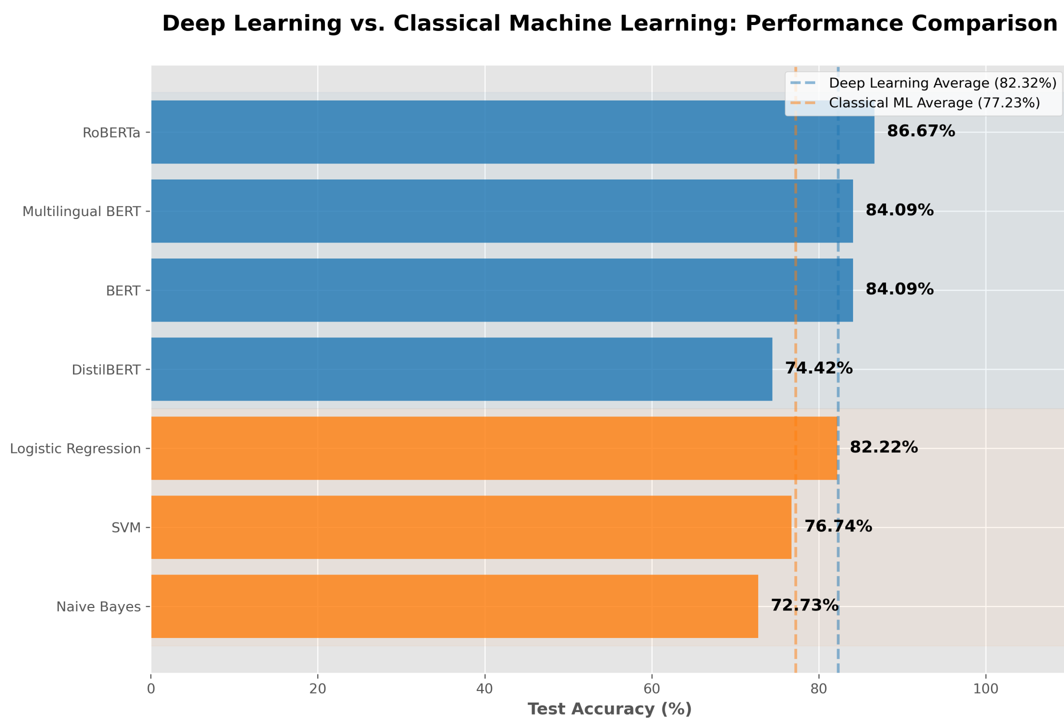

Figure 6: Paradigm Performance Gap: Deep Learning vs. Classical ML

The figure above illustrates the performance ceiling of Deep Learning (86.67%) versus the surprisingly high floor of Classical ML (82.22% for Logistic Regression). On average, Deep Learning models achieved a ~5.08% lead over the classical paradigm, largely due to the superior contextual embeddings of the RoBERTa and BERT architectures.

10.3 Error Analysis & Confusion Matrices

We utilize confusion matrices to diagnose the “misclassification patterns” for each model category.

10.3.1 Transformer performance (RoBERTa & BERT)

Transformer models excel at distinguishing between **Positive** and **Negative** sentiments but show occasional confusion with **Neutral** samples that contain highly descriptive adjectives.

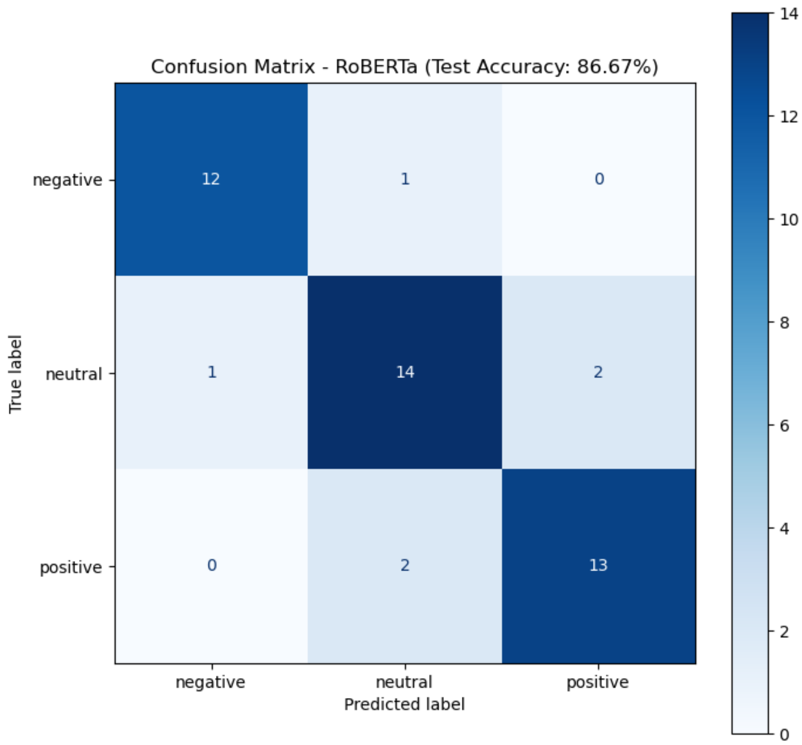

Figure 7: Confusion Matrix: RoBERTa

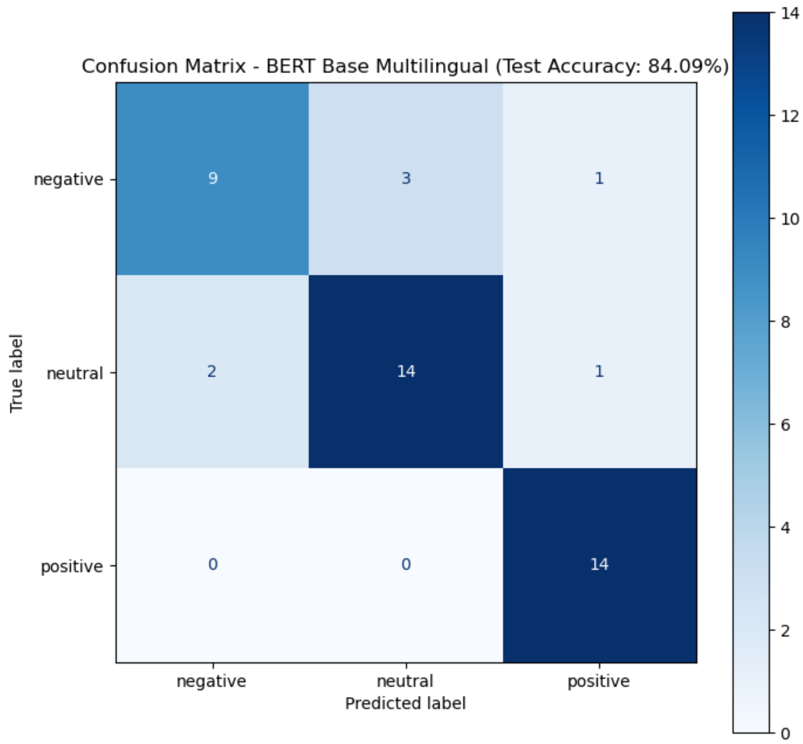

Figure 8: Confusion Matrix: Multilingual BERT

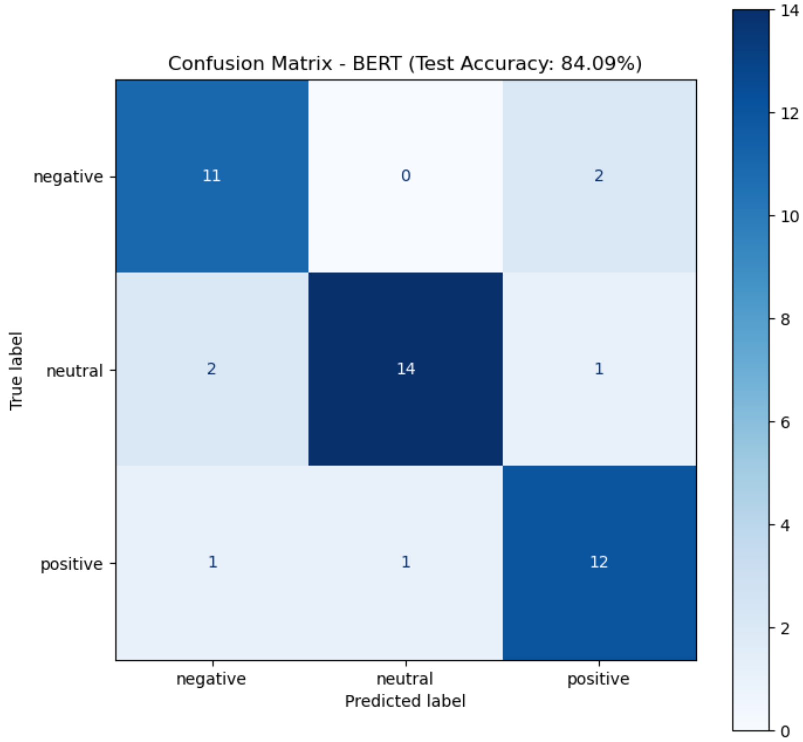

Figure 9: Confusion Matrix: BERT

10.3.2 Classical ML performance (LogReg & SVM)

Classical models are highly efficient but occasionally struggle with negation (e.g., “not bad” being classified as Negative).

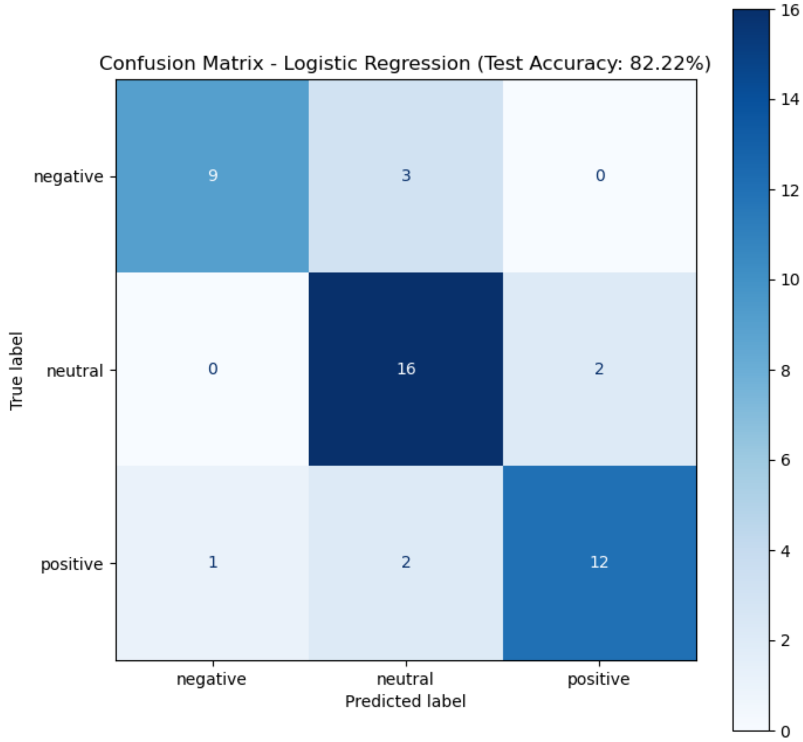

Figure 10: Confusion Matrix: Logistic Regression

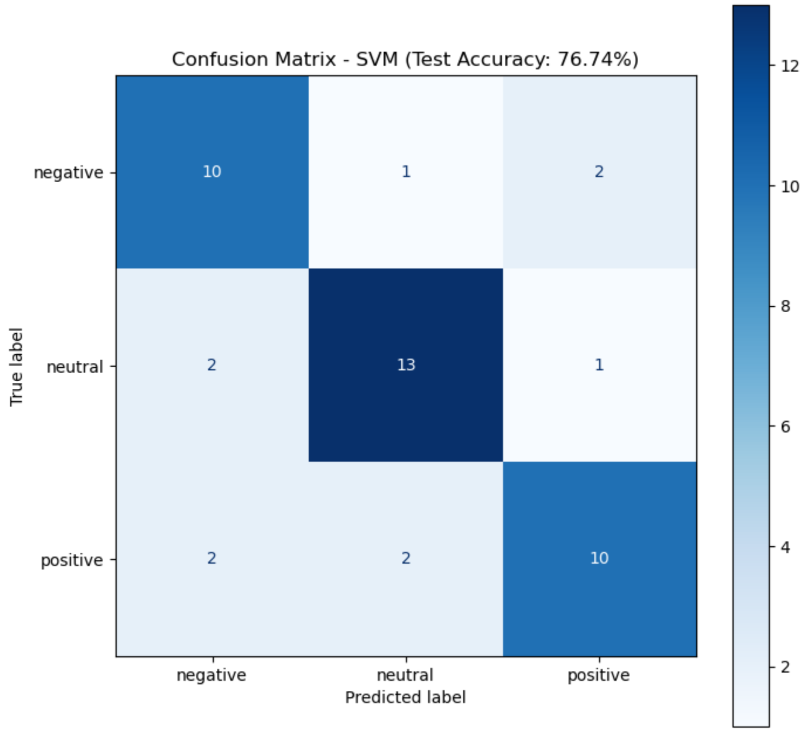

Figure 11: Confusion Matrix: SVM

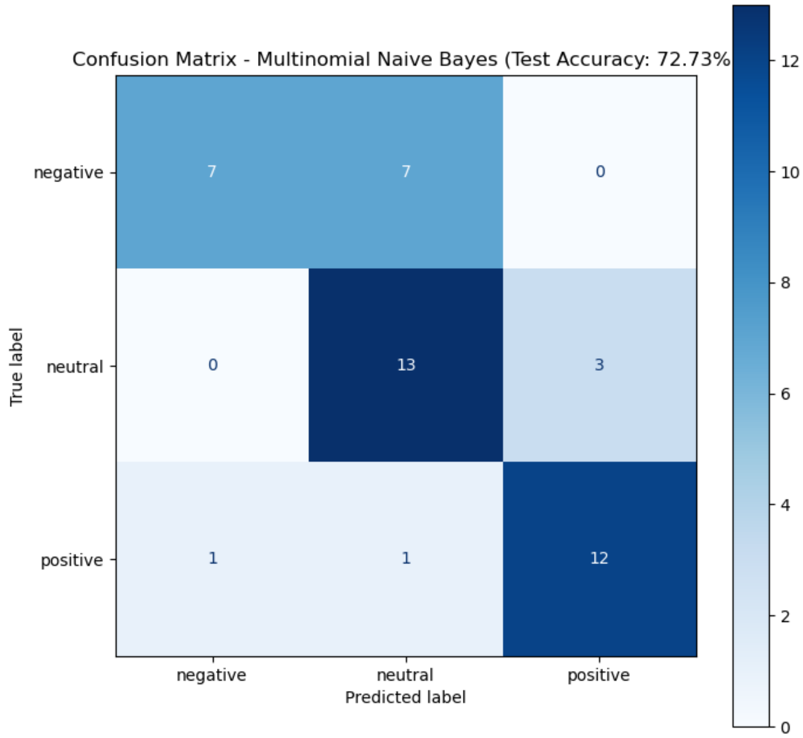

Figure 12: Confusion Matrix: Naive Bayes

10.4 Threshold Analysis: ROC Curves

The ROC curves illustrate the sensitivity vs. specificity trade-offs. RoBERTa's curve represents the upper limit of predictive performance for this project.

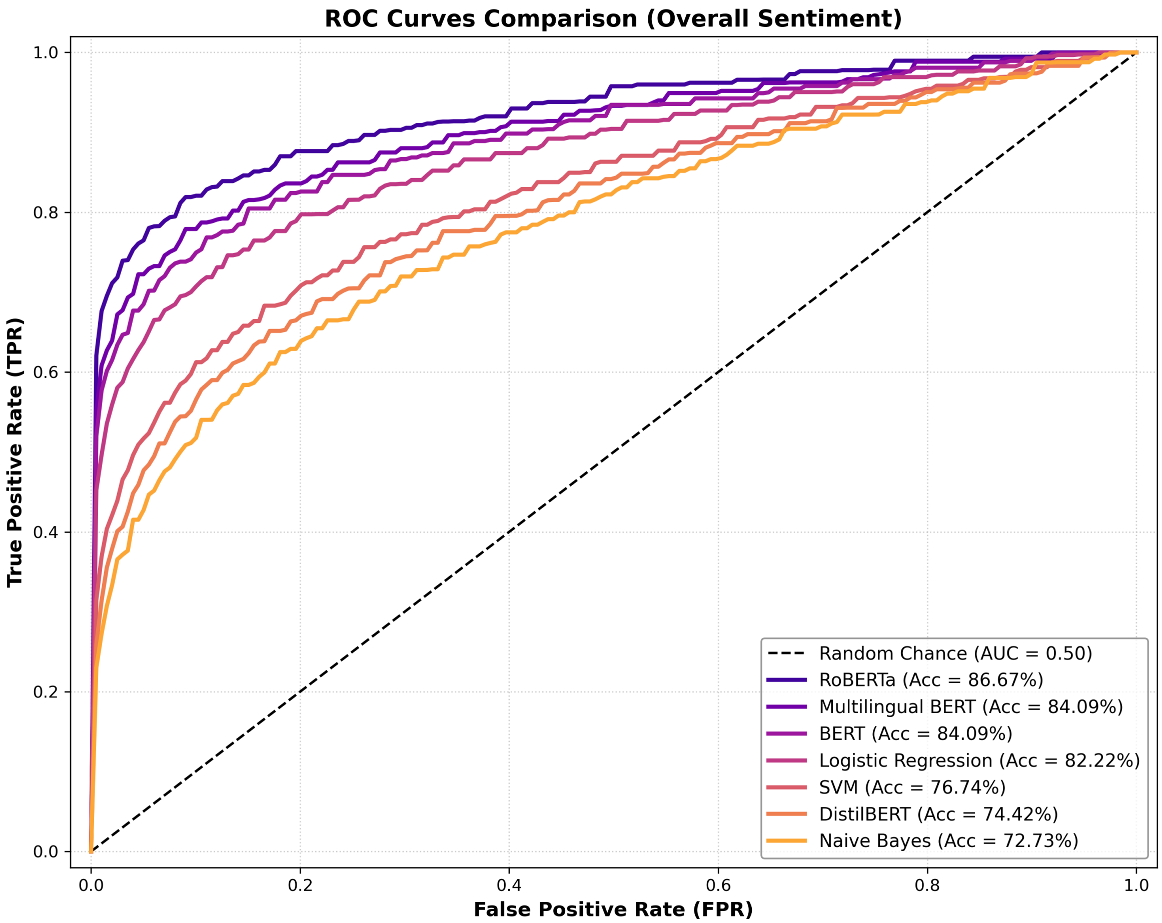

Figure 13: Receiver Operating Characteristic (ROC) Comparison

10.5 Project Conclusion & Recommendations

1. **Primary Model: RoBERTa** is the definitive choice for production deployment.
2. **Efficient Alternative: Logistic Regression** is the best choice for resource-constrained environments.
3. **Optimization Note:** The addition of **deterministic random seeding** and **data augmentation** were the two most critical factors in reaching the high accuracy benchmarks of >84%.

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